

```

tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ gedit practical12.c

** (gedit:3557): WARNING **: 03:24:58.665: Could not load Peas repository:
ypelib file for namespace 'Peas', version '1.0' not found

** (gedit:3557): WARNING **: 03:24:58.666: Could not load PeasGtk repositio
: Typelib file for namespace 'PeasGtk', version '1.0' not found
^C
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ gcc -Wall -Wextra -pthr
ad practical12.c -o practical12
practical12.c: In function 'deadlock_thread1':
practical12.c:300:11: warning: unused parameter 'arg' [-Wunused-parameter]
  300 |     void *arg
      |     ~~~~~^~
practical12.c: In function 'deadlock_thread2':
practical12.c:345:11: warning: unused parameter 'arg' [-Wunused-parameter]
  345 |     void *arg
      |     ~~~~~^~
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ./practical12
=====
                PRACTICAL 12
=====

Commands:
pc2          - Producer-consumer buffer size 2
pc5          - Producer-consumer buffer size 5
pc10         - Producer-consumer buffer size 10
throughput   - Compare buffer sizes
deadlock     - Demonstrate deadlock
prevent      - Demonstrate prevention
all          - Run producer-consumer + prevention
exit         - Exit

practical12> pc2

=====
    PRODUCER-CONSUMER TEST
=====
Buffer size: 2
Items       : 20

Producer 1 produced 1000

```

```
Producer 1 produced 1000
Consumer 1 consumed 1000
Producer 1 produced 1001
Consumer 1 consumed 1001
Producer 1 produced 1002
Producer 1 produced 1003
Consumer 1 consumed 1002
Producer 1 produced 1004
Consumer 1 consumed 1003
Producer 1 produced 1005
Consumer 1 consumed 1004
Producer 1 produced 1006
Consumer 1 consumed 1005
Producer 1 produced 1007
Consumer 1 consumed 1006
Producer 1 produced 1008
Consumer 1 consumed 1007
Producer 1 produced 1009
Consumer 1 consumed 1008
Producer 1 produced 1010
Consumer 1 consumed 1009
Producer 1 produced 1011
Consumer 1 consumed 1010
Producer 1 produced 1012
Consumer 1 consumed 1011
Producer 1 produced 1013
Consumer 1 consumed 1012
Producer 1 produced 1014
Consumer 1 consumed 1013
Producer 1 produced 1015
Consumer 1 consumed 1014
Producer 1 produced 1016
Consumer 1 consumed 1015
Producer 1 produced 1017
Consumer 1 consumed 1016
Producer 1 produced 1018
Consumer 1 consumed 1017
Producer 1 produced 1019
Consumer 1 consumed 1018
Consumer 1 consumed 1019

Producer threads completed.
```

```
=====
PRODUCER-CONSUMER TEST
=====
```

```
Buffer size: 10
Items       : 20
```

```
Producer 1 produced 1000
Consumer 1 consumed 1000
Producer 1 produced 1001
Consumer 1 consumed 1001
Producer 1 produced 1002
Consumer 1 consumed 1002
Producer 1 produced 1003
Producer 1 produced 1004
Consumer 1 consumed 1003
Producer 1 produced 1005
Consumer 1 consumed 1004
Producer 1 produced 1006
Producer 1 produced 1007
Consumer 1 consumed 1005
Producer 1 produced 1008
Consumer 1 consumed 1006
Producer 1 produced 1009
Producer 1 produced 1010
Consumer 1 consumed 1007
Producer 1 produced 1011
Consumer 1 consumed 1008
Producer 1 produced 1012
Consumer 1 consumed 1009
Producer 1 produced 1013
Producer 1 produced 1014
Consumer 1 consumed 1010
Producer 1 produced 1015
Consumer 1 consumed 1011
Producer 1 produced 1016
Producer 1 produced 1017
Consumer 1 consumed 1012
Producer 1 produced 1018
Consumer 1 consumed 1013
Producer 1 produced 1019
Consumer 1 consumed 1014
```

```
Consumer threads completed.
Produced : 20
Consumed : 20
CPU time : 0.003116 seconds
Synchronization correctness: PASS
practical12> throughput
```

```
=====
BUFFER SIZE COMPARISON
=====
```

```
Testing different buffer sizes.
Larger buffers generally allow more
temporary producer/consumer imbalance
```

```
=====
PRODUCER-CONSUMER TEST
=====
```

```
Buffer size: 2
Items       : 20
```

```
Producer 1 produced 1000
Consumer 1 consumed 1000
Producer 1 produced 1001
Consumer 1 consumed 1001
Producer 1 produced 1002
Consumer 1 consumed 1002
Producer 1 produced 1003
Producer 1 produced 1004
Consumer 1 consumed 1003
Producer 1 produced 1005
Consumer 1 consumed 1004
Producer 1 produced 1006
Consumer 1 consumed 1005
Producer 1 produced 1007
Consumer 1 consumed 1006
Producer 1 produced 1008
Consumer 1 consumed 1007
Producer 1 produced 1009
Consumer 1 consumed 1008
Producer 1 produced 1010
Consumer 1 consumed 1009
Producer 1 produced 1011
Consumer 1 consumed 1010
```

```

=====
DEADLOCK SCENARIO
=====
Thread 1 requires A -> B
Thread 2 requires B -> A

Four necessary deadlock conditions:
1. Mutual exclusion
2. Hold and wait
3. No preemption
4. Circular wait

Demonstrating resource dependency:

Thread 1: locking Resource A...
Thread 1: Resource A acquired.

Thread 2: locking Resource B...
Thread 2: Resource B acquired.
Thread 1: waiting for Resource B...
Thread 2: waiting for Resource A...

DEADLOCK CONDITION OBSERVED.
Thread 1 holds A and waits for B.
Thread 2 holds B and waits for A.
Neither thread can proceed.
prevent

^C
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ./practical12
=====
PRACTICAL 12
=====

Commands:
pc2      - Producer-consumer buffer size 2
pc5      - Producer-consumer buffer size 5
pc10     - Producer-consumer buffer size 10
throughput - Compare buffer sizes
deadlock - Demonstrate deadlock
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all      - Run producer-consumer + prevention
exit     - Exit

```

```
practical12> prevent
```

```
=====
DEADLOCK PREVENTION
=====
```

```
Resource ordering strategy:
Every thread must acquire A before B.
This prevents circular wait.
```

```
Safe thread 1 requesting resources...
Resource A acquired.
Resource B acquired.
Safe thread 1 using resources.
```

```
Safe thread 2 requesting resources...
Safe thread 1 released resources.
Resource A acquired.
Resource B acquired.
Safe thread 2 using resources.
Safe thread 2 released resources.
```

```
Deadlock prevention result: PASS
Both threads completed without deadlock.
practical12>
```